Hyeonmin Jeon

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Education

Yonsei University Seoul, Korea
M.S. in Industrial Engineering, GPA 3.97/4.0 *Mar 2025 – Feb 2027 (expected)*

- Advisor: Prof. Kyong Joo Oh, Financial Investment AI Lab.
- Coursework: Stochastic Modelling Theory, Probabilistic Models & Bayesian Inference, Time-Series Analysis, Deep Learning Theory, Advanced Investment Engineering, Financial Engineering, Corporate Finance.

Chung-Ang University Seoul, Korea
B.S. in Electrical and Electronics Engineering, GPA 3.59/4.0 *Mar 2019 – Feb 2025*

- Thesis: *Light-weighting Tiny YOLOv3: The Impact of Quantization on Real-World Applications* (advisor Prof. Minhyeok Lee).
- Coursework: Linear Algebra, Numerical Analysis, Random Variables, Signals & Systems, Machine Learning, Algorithms.

Research Interests

Simulation optimization under input uncertainty; generative models (diffusion, normalizing flows) as simulation input models; learned surrogates; sequential decision-making with statistical guarantees.

Publications and Working Papers

* corresponding author.

Journal manuscripts under review

1. **H. M. Jeon**, H. S. Lee*, K. J. Oh*. Conditional Diffusion Transformers for Multi-Asset Scenario Generation and Portfolio Risk Measurement. *Expert Systems with Applications*, under review, 2026.
2. **H. M. Jeon**, H. S. Lee*, K. J. Oh*. FinPhasor: Phase-Preserving Measurement of Cross-Sectional Timing in Equity Markets. *Expert Systems with Applications*, under review, 2026.
3. S. H. Kim, **H. M. Jeon**, D. H. You, J. Jeon, S. Baek, H. S. Lee, K. J. Oh*. Deep Convolutional Factor Prediction with Explainable AI for Enhanced Investment Strategy. In preparation for resubmission, 2026.
4. D. H. You, J. Jeon, **H. M. Jeon**, S. H. Kim, S. H. Yoo, T. Y. Kim, H. S. Lee, K. J. Oh*. Benchmark-Aware Enhanced Indexing Guided by the Volume–Price Index. In preparation for resubmission, 2026.

Working papers

5. **H. M. Jeon**. Learned Surrogate Priors for Bayesian Model Selection: Density, Uncertainty, and the Cost of a Sampler. Lab seminar, Jan 2026. [link]
6. **H. M. Jeon**. Bayesian Capacity Sizing for an Exchange Matching Queue under Over-Dispersed Arrivals. 2026. [link]

Conference presentations

7. Y. Jeong*, H. Kim, D. H. You, **H. M. Jeon**, A. de Castro, T. Lee. A Study for Forecasting Solar Energy Power Generation. *45th International Symposium on Forecasting*, Beijing, Jul 2025. Talk; joint work with SAS Institute.
8. **H. M. Jeon**, M. Lee. Dealing with the Imbalanced-Data Problem with a Diffusion Model. *KICS Fall Conference*, Nov 2023.
9. **H. M. Jeon** et al. Personalized Sleep Score Prediction Model Using a Lifelog Dataset. *ETRI Human Understanding AI Paper Challenge*, 2024.

Research Experience

Financial Investment AI Lab, Yonsei University Seoul, Korea
Graduate Research Assistant *Mar 2025 – Present*

- Conditional diffusion transformer for joint multi-asset return scenarios. Evaluated by the VaR/ES reserves it sets, not by sample realism.
- Cross-sectional timing from complex wavelet scalograms. Phase is preserved so that when each stock reprices

is used for ranking.
- Normalizing-flow prior for Bayesian factor selection. Matches the exact posterior over models (TV 0.016); tuned by Bayesian optimization; compared with a density-free GAN prior.
- Ranking and selection under input uncertainty. Implemented and verified KN, feasibility determination, recycled observations and CRN; measured how a fitted input model breaks the selection guarantee.

TIPS R&D Project with Urban Complex Seoul, Korea
Researcher, artist-style generative engine *Mar 2025 – Aug 2026*

- QLoRA style classifier on Qwen2-VL-7B with chain-of-thought captions: 91.4% seen / 76.0% unseen; 96.0% fused with a DINOv2 prior.
- SDXL generation with domain checkpoints routed by the classifier; style memory module.

GAIL (AI Lab), Chung-Ang University Seoul, Korea
Undergraduate Researcher *Dec 2022 – Dec 2024*

- Diffusion oversampling for imbalanced fraud data; latent-diffusion portfolio study; lifelog sleep-score prediction; 3D point-cloud classification.

Carbon-Neutral ESG Research Group, Chung-Ang University Seoul, Korea
Team Lead, Green-Navi (Green AI track) *Apr 2023 – Dec 2023*

- Reinforcement learning for fuel-optimal urban routing in AIMSUN. Presented at the research fair, Nov 2023 (MSIT/NRF programme).

Selected Projects

Retina Exam Room Scheduling, Nune Eye Hospital
Discrete-event simulation and reinforcement learning *Feb 2026 – Present*

- Recalibrated a simulator to hospital records. Compared rule, behaviour-cloned and PPO policies on a 44-day replay: time in room 12.2 → 8.2 min. Read-only EMR integration.

Simulation-Methodology Reimplementations
rs-lab, Input-Uncertainty Select, TailFlow, Eco-Route RL, roughvol-lab *Sep 2026*

- Ranking & selection with verified guarantees; input-uncertainty effects on selection; conditional diffusion as a risk engine; CO_2-optimal routing on SUMO (P(correct selection) 0.956); rough volatility.

Lucubra
LLM agents for paper verification and referee reports *Sep 2026 – Present*

Light-weighting Tiny YOLOv3
Graduation thesis, team of three *2024*

- Int8 quantization with layer fusion. Footprint 1.6 GB → 100 MB; inference 22–64% faster on five devices; mAP −2.78 pt on PASCAL VOC.

Teaching

Yonsei University Seoul, Korea
Teaching Assistant, Graduate Program in AI Investment Engineering *Sep 2025 – Present*

Yonsei University Seoul, Korea
Teaching Assistant, Financial Engineering (undergraduate) *Spring 2026*

Awards

Special Prize (data use), Chungcheong ICT Innovation Public-Data Competition *2024*

Skills

Programming	Python (NumPy, SciPy, pandas, PyTorch, CUDA), C++, MATLAB, TypeScript, SQL, LaTeX
Methods	Discrete-event and Monte Carlo simulation, ranking & selection, variance reduction, diffusion and flow models, Bayesian inference, reinforcement learning, time-series analysis
Tools	Eclipse SUMO, AIMSUN, SAS, Hugging Face, Linux multi-GPU servers
Languages	Korean (native), English (working proficiency)

Updated October 17, 2026